

Calculating the Moment of Inertia of the Maxwell Wheel

Introduction

In the origin of this experiment, the drive arose from a video showing the dynamics of a Maxwell wheel/pendulum system (Orgill (2022)). The video illustrated a compelling phenomenon: when the wheel was set into motion, the entire system exhibited a decrease in weight on a scale. The interaction between rotational motion and gravitational forces prompted intrigue into the subject, ultimately driving the conception of an experimental investigation.

The initial intention was to quantify the observed decrease in the weight of the system as the wheel rotated. With the setup of the experiment, the methodology was more than undesirable, bringing uncertainty and variables that ultimately rendered the data useless. Due to the lack of relevant measuring devices (a sufficiently large scale), a change was opted instead, with each string possessing a dynamometer/neutronmeter, which created an exceedingly large sway in the force. Faced with the constraints of available materials and the lack of precision, a pivotal decision was made to redirect the focus of the experiment. The recalibration aimed to clarify the relations observed in the measured variables (PHYWE (1998)). The experiment was shifted to calculate the moment of inertia of the Maxwell wheel by building a hypothetical formula for the position of the wheel after a given amount of time.

The interest in mechanics and the intrinsic appeal of comprehending the fundamental physics behind these occurrences were crucial in guiding this experimental path. Within the domain of scientific investigation, there is a deep fascination with exploring the relationship between rotational movement and gravitational forces. This has sparked a strong curiosity in solving the challenges associated with the Maxwell wheel.

By releasing the wheel from varying heights and measuring the time it takes to reach a designated point without making contact with the ground, the experiment aims to create a connection between the observed phenomena and the physical principles governing the system. The emphasis lies not only on ascertaining the moment of inertia, but also on unveiling the relationships between height, time, and the inherent constants in Maxwell wheel dynamics. This experiment shows the drive to explore a physical system and create an experiment within it even when faced with inaccessible measuring devices.

Theory

The motion of the Maxwell wheel is described by fundamental equations governing displacement, velocity, angular velocity, and angular acceleration within the system. These equations establish a framework for understanding the dynamics of the rotating wheel.

First, it is helpful to describe the system itself. A Maxwell wheel is an object named after James Clerk Maxwell (the exact history of the object is blurry at best, with the first mention of its name being the Maxwell wheel cited in (Germany FR. National Commission for UNESCO, 1974)). It consists of a disc or a wheel with an axle that passes through it. At both ends of the axle, two strings are attached. This is commonly used to teach about the conservation of energy. The way it does so is by winding the wheel and attaching the ends of the strings in such a way that the axle is level. The wheel is then let go and set in motion.

Visualization of the system is aided by a diagram (Figure 1) in order to help us visualize what it is that we are trying to describe.

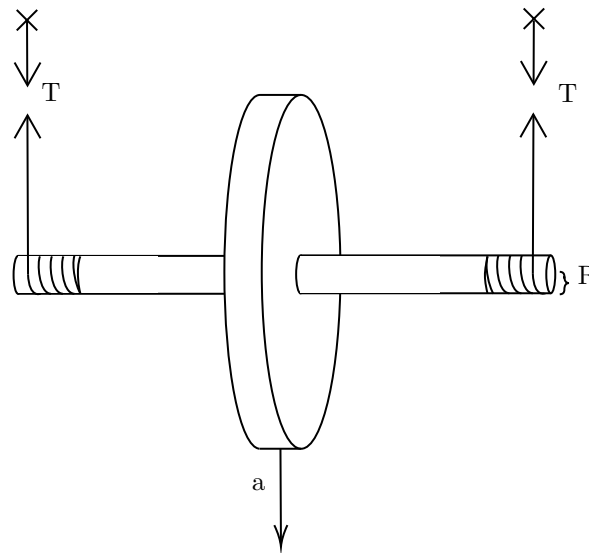


Figure 1: Diagram of the experiment

Through this diagram, we can see how the Maxwell wheel experiment is set up. Although not described, an overall downward force mg can also be considered. It represents the gravitational acceleration constant multiplied by the mass of the wheel. For simplicity's sake, consider the rope to be weightless and attached to a fixed point. Air resistance is taken as null. With these preconceptions, the description of the system can be initiated (Chintalapudi (2008)). Through the torque equation, an association of τ with $I\alpha$ can be made (Atkins et al. (2019)). In this case (and throughout this paper), τ represents torque, while I and α represent moment of inertia (a quantity that represents resistance to angular acceleration of a body) and angular acceleration. The other equation for the torque ($\tau = Fs \cdot \sin \Theta$) creates another relation for the torque to the tension of the string, and the radius of the spindle can be made. Here, we can assume that the angle is always 90 degrees and, therefore, $\sin \Theta$ is equal to one. This happens

because any tangent to a circle is perpendicular to the line created by the tangent point and the center of the circle (the radius). The radius used for the torque equation is the radius of the axle, since it is the point in which there is tension. For clarity, the radius of the axle will be referred to as R . The equation created can be written as:

$$\tau = I\alpha$$

$$2TR = I\alpha$$

Considering a non-slipping string, a relation between angular speed to linear speed can be created. An illustration of such a relation can be seen in the following diagram (Figure 2), holding true only while the string does not experience any change in distance from the rotational axis or any slippage (Cummings et al. (2004); PHYWE (1998)).

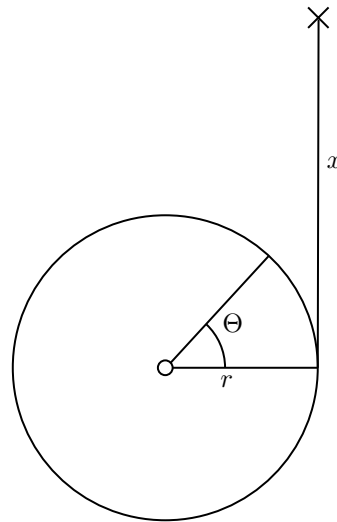


Figure 2: Diagram of the relationship between angular speed and linear speed

Here, $d\vec{x} = d\vec{\theta} \times d\vec{r}$ allows for the formulation of further equations with the preliminary knowledge of circular motion (Bowen-Jones and Homer (2014)).

$$\begin{aligned}d\vec{x} &= d\vec{\theta} \times d\vec{r} \\ \vec{v} \equiv \frac{d\vec{x}}{dt} &= \frac{d\vec{\theta}}{dt} \times \vec{r} \equiv \vec{\omega} \times \vec{r} \\ \omega &= \frac{v}{r} \\ \alpha &= \frac{a}{r}\end{aligned}$$

Referring back to the Maxwell experiment equation, replacing α with $\frac{a}{r}$, and knowing that

the only force present is gravity, a further expression of the system can be made. Equating $2T$ with mg , where m is the mass of the Maxwell wheel, and g is the gravitational constant. This is taken from the fact that both $I\alpha$ and mgR are related to the tension in the strings (or $2T$).

$$\begin{aligned}\tau &= I\alpha \\ 2TR &= I\frac{a}{R} \\ mgR &= I\frac{a}{R}\end{aligned}$$

Solving for a produces (Chintalapudi (2008)):

$$\begin{aligned}mgR^2 &= Ia \\ \frac{mgR^2}{I} &= a\end{aligned}$$

This, although true for a static observation, is not correct when the wheel is in motion. These equations, although helpful, describe only the acceleration of the wheel while it is not already moving.

The Maxwell wheel experiment takes potential energy and turns it into kinetic energy, with the kinetic energy being composed of translational (linear from the center of mass) or rotational (Whittaker (1988)). The important distinction is that a perfect system does not experience a loss of energy. For that reason, we can consider the sum of energies of the system to be constant (depicted as C in the following equation) and, therefore, the derivative of such to be equal to zero. The equation follows:

$$\begin{aligned}E_T &= C \\ \frac{d}{dt}(E_{sys}) &= 0 \\ \frac{d}{dt}(E_p + E_t + E_r) &= 0 \\ \frac{d}{dt}\left[mgh(t) + \frac{1}{2}mv^2(t) + \frac{1}{2}I\omega^2(t)\right] &= 0\end{aligned}$$

Using the aforementioned notion of the equation between ω and $\frac{v}{r}$ (), a continuation of

the equation is possible.

$$\begin{aligned}\frac{d}{dt} \left[mgh(t) + \frac{1}{2}mv^2(t) + \frac{1}{2}I \left(\frac{v}{R} \right)^2(t) \right] &= 0 \\ \frac{d}{dt} [mgh(t)] + \frac{d}{dt} \left[\frac{1}{2}mv^2(t) \right] + \frac{d}{dt} \left[\frac{1}{2}I \frac{v^2(t)}{R^2} \right] &= 0\end{aligned}$$

Knowing that the mass, moment of inertia, gravitational constant, and radius of the axle are constants, we can isolate them from their respective derivations since they are multiplying constants.

$$mg \cdot \left\{ \frac{d}{dt} [h(t)] \right\} + \frac{1}{2}m \cdot \left\{ \frac{d}{dt} [v^2(t)] \right\} + \frac{I}{2R^2} \cdot \left\{ \frac{d}{dt} [v^2(t)] \right\} = 0$$

Taking $\frac{d^2(a(t))}{dt^2} = \frac{d(v(t))}{dt} = h(t)$ to be true, we can proceed with the derivation while using the chain rule, where applicable.

$$\begin{aligned}mg(v) + \frac{1}{2}m(2va) + \frac{I}{2R^2}(2av) &= 0 \\ mgv + mva + \frac{Iav}{R^2} &= 0\end{aligned}$$

We can isolate v and divide both sides. This essentially removes v from the current equation.

$$\begin{aligned}v \cdot \left(mg + ma + \frac{Ia}{R^2} \right) &= 0 \\ mg + ma + \frac{Ia}{R^2} &= 0\end{aligned}$$

We are then able to isolate the acceleration (a) through common arithmetic means, such as factorization and division.

$$ma + \frac{Ia}{R^2} = -mg$$

$$\begin{aligned}a \cdot \left(m + \frac{I}{R^2}\right) &= -mg \\a \cdot \left(\frac{mR^2 + I}{R^2}\right) &= -mg \\a &= \left(\frac{R^2}{mR^2 + I}\right) \cdot -mg \\a &= -\frac{mgR^2}{mR^2 + I}\end{aligned}$$

In order to get the desired relation between height/position and time squared, we need to integrate this equation twice in order to get position from acceleration. As we already recognized earlier, this only holds true while we consider acceleration, speed, and position to be related through derivation and integration. The other side of the equation is constant throughout time and is therefore a straightforward operation since it is a multiplying constant that can be isolated.

$$\begin{aligned}\int \left(\int a \, dt\right) dt &= \int \left(\int -\frac{mgR^2}{mR^2 + I} \, dt\right) dt \\ \int \left(\int a \, dt\right) dt &= -\frac{mgR^2}{mR^2 + I} \cdot \int \left(\int 1 \, dt\right) dt \\ \int v + v_0 \, dt &= -\frac{mgR^2}{mR^2 + I} \cdot \int t \, dt \\ h(t) + v_0 t + h_0 &= -\frac{mgR^2}{mR^2 + I} \cdot \frac{1}{2} t^2 \\ h(t) &= -\frac{1}{2} \cdot \frac{mgR^2}{mR^2 + I} t^2 - v_0 t - h_0\end{aligned}$$

This gives us the desired equation, however, the moment of inertia I used is not the "symmetrical" inertia I_s (Beginners' practical course in physics, Department of Physics, University of Basel (2021)). This could theoretically be calculated through Steiner's Theorem:

$$I = I_s + mr^2$$

Although this would work, we have neither the means necessary to calculate r , nor do we have the capability to understand/explain Steiner's theorem. This, however, is not a big issue, as our setup provides a meaningless difference between I and I_s given the uncertainty.

Methodology

For this experiments, our independent variable is height, where we adjust it throughout to have various data sets (to reduce random error). Our dependent variable is the time it takes for the wheel to fall to the given height. We aim to measure the time it takes to fall in order to obtain the controlled variable, the moment of inertia, as it is constant throughout the experiment.

For the experimental setup, the Maxwell wheel is suspended in midair, accompanied by the presence of a meter or calibration stick positioned adjacent to it. To ensure accurate data collection, a smartphone is placed on the side of the contraption. The phone is oriented to capture both the side profile of the wheel and the meter, maintaining an equal distance between these components and the camera lens.

The objective of this setup is to record and analyze the motion of the center of mass of the Maxwell wheel. The process involved filming the downward motion of the axle while keeping track of its position over time. The meter is included in the field of view, serving as a better reference scale (better than using the spindle as its own reference, as the bigger calibration stick introduces a smaller uncertainty). To further decrease the uncertainty, an extra bar is clamped between the two stands to decrease the lateral vibrations created by falling weight.

In addition to the setup for measuring the position through time of the Maxwell wheel, a scale is also used in order to measure the weight of the wheel at rest, and a caliper to measure the diameter of the axle.

To facilitate the data collection process, the "Tracker" program is used (Douglas Brown (2024)). This software allows tracking of objects in motion from recorded video footage. The recorded data, which include both the temporal aspect and the corresponding distances covered by the wheel spindle, are subsequently extracted. Below is a photo of the experimental arrangement taken from the view of the recording camera (Figure 3):



Figure 3: Experimental Setup

As seen in the photo, the materials necessary are as follows:

- Two support bases
- Three support rods
- Four right-angled clamps
- Calibration stick
- Maxwell wheel
- Digital balance ($\pm 0.001\text{kg}$)
- Caliper ($\pm 0.00005\text{m}$)
- String
- Camera stand
- Meter scale (1 meter)

The experiments are then recorded with the wheel being "rolled up" to different heights and recorded with the camera. Three main videos are made with 5-14 different runs being made. An extra run is made with the weight being "rolled up" to its maximum height and being released without being interfered with until it completes many oscillations. This last run is made to evaluate whether there are many losses of energy in the system.

This setup aimed to provide a detailed understanding of the wheel's descent dynamics and facilitate the extraction of valuable quantitative data for subsequent analysis.

Data

The "common" data can be laid on a table as such:

	Weight (kg)	Diameter of axle (m)
The Maxwell wheel	1.012	0.013
Uncertainty of the measurement	0.001	0.00005

Data extraction from the Tracker program is primarily focused on capturing the vertical position of the Maxwell wheel spindle over time. Four different videos were recorded, each having a different number of trials. These are titled "x-y", with the x-value representing the video and the y-value representing the trial number. These are completely arbitrary and used solely for ease of processing (the third video was the one used to measure the resistance present in the system in this instance). These vertical position data were crucial for understanding the dynamics of the wheel's descent. The extracted data can be seen in figure 4.

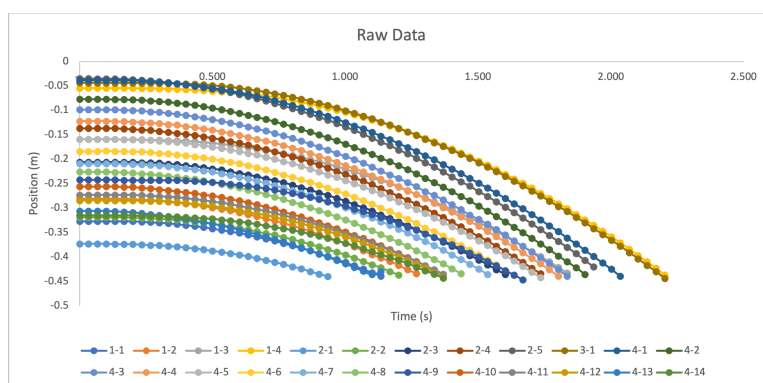


Figure 4: Vertical Position vs Time

The third trial can also be graphed as such (Figure 5):

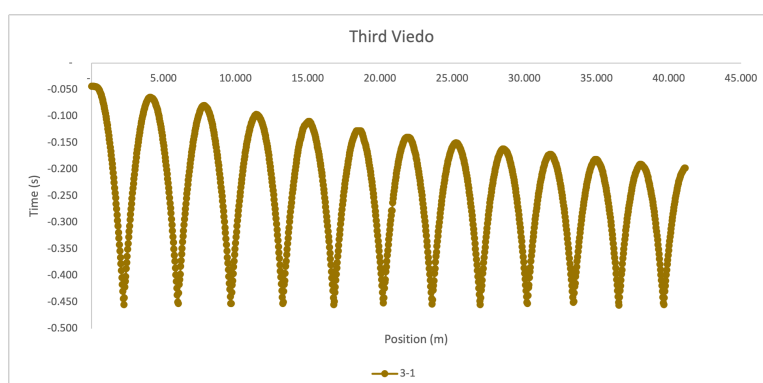


Figure 5: Vertical Position vs Time of the third video

The third video is useful to demonstrate that, although a considerable amount of resistance in the system is present, it is not overwhelming and does not massively impact the obtained data. It succeeds in demonstrating that the system does have uncertainties that come from

physical constraints such as air friction.

When the raw data were obtained, further adjustments were made to enhance its clarity and relevance. The first significant modification involved setting the point of the axle at zero meters in time. This adjustment allowed for a more meaningful representation of the vertical position, highlighting the relative changes in height as the wheel descended (Figure 6).

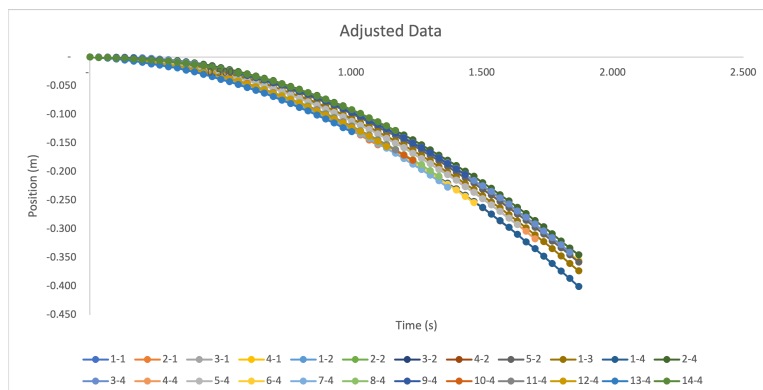


Figure 6: Adjusted Vertical Position vs Time

Subsequently, the processed data were graphed to visualize the relationship between time and vertical position. The initial graph provided information on the general descent pattern of the wheel. However, to explore potential connections between time and acceleration due to gravity, the data were graphed for the third time, with time squared (Figure 7).

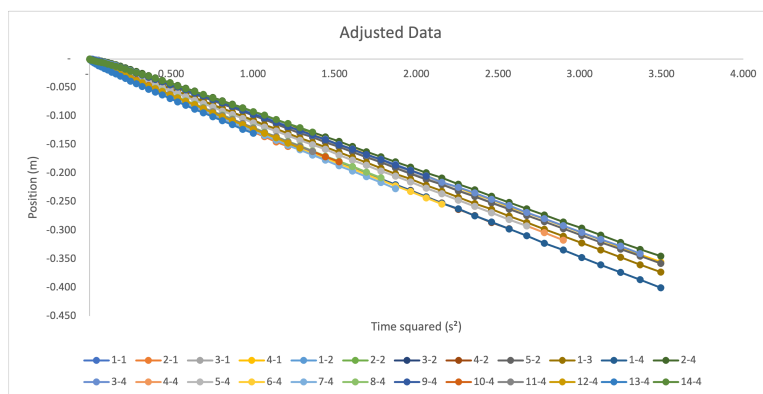


Figure 7: Adjusted Vertical Position Vertical Position vs Time Squared

These graphical representations form the basis for our quantitative analysis, allowing us to discern patterns and draw meaningful conclusions regarding the experimental outcomes.

Analysis

The analysis of the Maxwell wheel experiment offers insight into the conservation of energy within the system. Examining the data of the third video (figure 5) reveals a notable absence of significant energy loss during the wheel's descent, aligning with the physics that emphasizes the conservation of mechanical energy within an isolated system.

Looking at this trend, an analysis of the experimental data was undertaken, leading to the creation of a trend line that includes all experimental trials. The method used for the junction of the data was the average of the positional data points for each given time. The chosen graph (Figure 8) plots the position against the squared time, offering a trendline on the relationship between time and the position that governs the Maxwell wheel. The uncertainty of the trendline has also been done through error bars and proceeded by the maximum and minimum trendlines within the error bars. The former have been created by calculating the standard deviation between the averaged points.

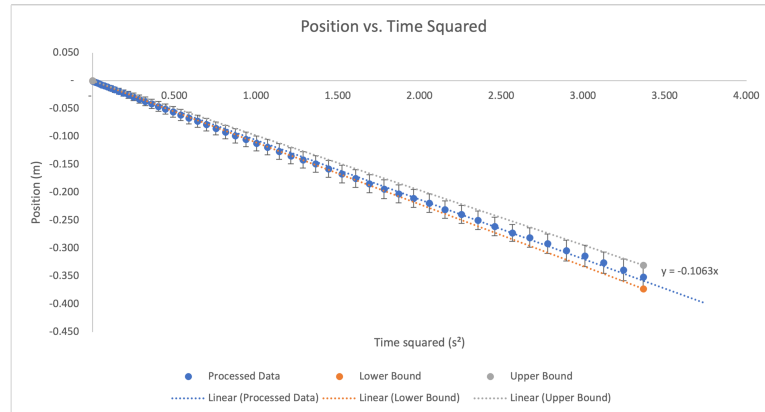


Figure 8: Trendline Analysis: Position vs. Time Squared

By fitting this trend line to the data, the priorly produced mathematical expression was used to derive the moment of inertia of the wheel.

The equation derived within the theoretical section is composed of $position(time) = time^2 \cdot constant$. Here, we can equate both constants together, as the trendline of the graph describes the previously derived constant. This will also be further simplified through simple arithmetic so that the moment of inertia is isolated. Most importantly, from the derived equation, both $v_0 t$ and h_0 can be taken as zero, since they represent the initial position and the initial speed. Since both of those have been set to 0, they can be ignored.

$$\begin{aligned}
 h(t) &= -\frac{1}{2} \cdot \frac{mgR^2}{mR^2 + I} t^2 - v_0 t - h_0 \\
 -\frac{1}{2} \cdot \frac{mgR^2}{mR^2 + I} &= \text{trendline} \\
 \frac{mgR^2}{mR^2 + I} &= -2 \cdot \text{trendline} \\
 \frac{mR^2 + I}{mgR^2} &= \frac{1}{-2 \cdot \text{trendline}} \\
 mR^2 + I &= \frac{mgR^2}{-2 \cdot \text{trendline}} \\
 I &= \frac{mgR^2}{-2 \cdot \text{trendline}} - mR^2
 \end{aligned}$$

Filling in the data, we receive the following:

$$\begin{aligned}I &= \frac{mgR^2}{-2 \cdot \text{trendline}} - mR^2 \\I &= \frac{1.012kg \cdot 9.81ms^{-2} \cdot (\frac{0.013m}{2})^2}{-2 \cdot -0.1063ms^{-2}} - 1.012kg \cdot (\frac{0.013m}{2})^2 \\I &= \frac{9.92772kg \cdot ms^{-2} \cdot 0.00004225m^2}{0.2126ms^{-2}} - 1.012kg \cdot 0.00004225m^2 \\I &= \frac{0.00041944617kg \cdot m^3s^{-2}}{0.2126ms^{-2}} - 0.000042757kg \cdot m^2 \\I &= 0.00197293588kg \cdot m^2 - 0.000042757kg \cdot m^2 \\I &= 0.00193017888kg \cdot m^2\end{aligned}$$

In order to obtain the uncertainty for this value, and therefore the amount of significant figures, we follow the rules of absolute and fractional uncertainty that follow the formula for the moment of inertia.

$$\Delta I = \left| \frac{mgR^2}{2 \cdot \text{trendline}} \cdot \left(\frac{\Delta m}{m} + 2 \left(\frac{\Delta R}{R} \right) + \left(\frac{\Delta \text{trendline}}{\text{trendline}} \right) \right) \right| + \left| mR^2 \cdot \left(\frac{\Delta m}{m} + 2 \left(\frac{\Delta R}{R} \right) \right) \right|$$

We proceed by substituting by the data.

$$\begin{aligned}\Delta I &= \left| 0.00197293588 \cdot \left(\frac{0.001}{1.012} + 2 \left(\frac{0.00005}{0.013} \right) + \left(\frac{0.006}{0.1063} \right) \right) \right| + \left| -0.000042757 \cdot \left(\frac{0.001}{1.012} + 2 \left(\frac{0.00005}{0.013} \right) \right) \right| \\ \Delta I &= |0.00197293588 \cdot (0.0651245)| + |-0.000042757 \cdot (0.00868045)| \\ \Delta I &= \pm 0.00012885761271771 \Delta I = \pm 0.0001\end{aligned}$$

With this uncertainty, we can finally say that the moment of inertia with its uncertainty that was calculated is the following:

$$I = 0.0019 \pm 0.0001kg \cdot m^2$$

This is the highlight of this paper. As the moment of inertia of the wheel cannot be simply measured, we can't prove this answer. However, this value corresponds in magnitude to the value we expect considering the weight of our wheel (Rodriguez (2023)).

Evaluation

The experimental methodology used in the investigation of the Maxwell wheel provided a comprehensive platform to study the dynamics of the system. The setup, which involved calibration,

precise measurements, and technological tools such as the Tracker program, aimed to minimize uncertainties and improve the reliability of the collected data. The strategic use of a calibration stick, camera, and additional stabilizing bar further underscored the commitment to experimentation.

However, some potential sources of error merit consideration. The assumption of negligible air resistance and a weightless rope, while simplifying the theoretical framework, may introduce minor deviations between the idealized model and real-world conditions. Additionally, the sensitivity of the experimental setup to external factors, such as lateral vibrations, underscores the importance of thorough control measures, though efforts to mitigate this were implemented. Having a heavier wheel might have helped reduce air resistance, while a stronger set-up could have diminished factors such as vibrations.

Conclusion

The analysis of the Maxwell wheel experiment provides compelling evidence to support the conservation of energy within the system. Trend-line analysis, graphing position against time squared, revealed a consistent pattern across various trials, indicating minimal energy loss during the wheel's descent.

Through the derived mathematical expression and subsequent calculations, the moment of inertia (I) for the Maxwell wheel was determined to be $0.0019 \pm 0.0001 \text{ kg m}^2$. This outcome aligns with rough theoretical expectations and comparisons (Rodriguez (2023); Serway (1986)), further validating the principles of energy conservation.

In conclusion, the Maxwell wheel experiment not only affirmed the principles of physics but also demonstrated the effectiveness of the experimental methodology in elucidating complex dynamics. The study contributes valuable information to the understanding of energy conservation in rotational motion systems.

References

- A.G. Atkins, M. Escudier, and T. Atkins. *A Dictionary of Mechanical Engineering*. Oxford Quick Reference. Oxford University Press, 2019. ISBN 9780198832102. URL <https://books.google.ch/books?id=iTSZwgEACAAJ>.
- Beginners' practical course in physics, Department of Physics, University of Basel, Mar 2021. URL <https://ap.physik.unibas.ch/PDF/Manuals/English/IM3en.pdf>.
- Michael Bowen-Jones and David Homer. *Oxford IB Diploma Programme: Physics Course Companion*. Oxford University Press-Children, 2014.
- Krishna Kant Chintalapudi. Irodov problem 1.244, Nov 2008. URL <https://irodovsolutionsmechanics.blogspot.com/2008/11/irodov-problem-1244.html>.
- K. Cummings, P.W. Laws, and E.F. Redish. *Understanding Physics, 1st Edition*. Recording for the Blind & Dyslexic, 2004. ISBN 9781118313299. URL <https://books.google.ch/books?id=iRocAAAAQBAJ>.
- Robert M Hanson Douglas Brown, Wolfgang Christian. Tracker (version 6.1.6), 2024. URL <https://physlets.org/tracker/>.
- Germany FR. National Commission for UNESCO and Christian-Albrechts-Universität Kiel (Germany FR). Institut für die Pädagogik der Naturwissenschaften. *Implementation of Curricula in Science Education: Report of an International Seminar on "the Implementation of Curricula in Science Education with Special Regard to the Teaching of Physics", Organized by the German Commission for UNESCO and the Institute for Science Education at the Univ. of Kiel, Kiel, March 16-18, 1972*. Deutsche Unesco-Kommission: Seminarbericht der Deutschen Unesco-Kommission. Pullach bei München : Verlag Dokumentation, 1974. ISBN 3-7940-5222-6. URL <https://unesdoc.unesco.org/ark:/48223/pf0000226888>.
- James Orgill. How does the anti-gravity wheel work?, Jul 2022. URL <https://www.youtube.com/watch?v=pwx12kwg0AM>.
- PHYWE. Mechanical conservation of energy / maxwell's wheel, December 1998. URL https://www.nikhef.nl/~h73/kn1c/praktikum/phywe/LEP/Experim/1_3_18.pdf. PHYWE series of publications.
- Sydney Rodriguez. Experiment 5: Maxwell's wheel, Sep 2023. URL <https://www.coursesidekick.com/physics/1669422>.
- Raymond A Serway. *Study Guide with Computer Exercises to Accompany Physics for Scientists and Engineers, [and] Physics for Scientists and Engineers with Modern Physics*. Saunders College Pub., 1986.

E. T. Whittaker. *A Treatise on the Analytical Dynamics of Particles and Rigid Bodies*. Cambridge Mathematical Library. Cambridge University Press, 1988.

Appendix

<i>Trials</i>	–	–	1					2					3	4														
<i>Runs</i>	<i>time</i>	<i>time squared</i>	1	2	3	4		1	2	3	4	5		1	1	2	3	4	5	6	7	8	9	10	11	12	13	14
<i>Data</i>	<i>t</i>	<i>t</i> ²	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	
	0	0	−0.327	−0.281	−0.159	−0.05473	−0.373	−0.319	−0.206	−0.137	−0.03515	−0.04388	−0.03871	−0.07687	−0.09857	−0.122	−0.159	−0.184	−0.209	−0.226	−0.242	−0.256	−0.273	−0.285	−0.306	−0.315		
	0.033	0.001089	−0.327	−0.281	−0.159	−0.05476	−0.373	−0.319	−0.206	−0.137	−0.0352	−0.0439	−0.03874	−0.07705	−0.0986	−0.122	−0.159	−0.184	−0.209	−0.226	−0.242	−0.256	−0.273	−0.284	−0.306	−0.315		
	0.067	0.004489	−0.327	−0.281	−0.159	−0.05475	−0.373	−0.319	−0.206	−0.137	−0.03522	−0.04392	−0.03867	−0.0772	−0.09858	−0.122	−0.159	−0.184	−0.209	−0.226	−0.242	−0.256	−0.273	−0.284	−0.306	−0.315		
	0.1	0.01	−0.327	−0.281	−0.159	−0.05483	−0.373	−0.319	−0.206	−0.137	−0.03538	−0.04393	−0.03872	−0.07727	−0.0988	−0.122	−0.159	−0.183	−0.209	−0.226	−0.242	−0.256	−0.273	−0.285	−0.306	−0.315		
	0.133	0.017689	−0.327	−0.281	−0.159	−0.05478	−0.374	−0.319	−0.206	−0.137	−0.03572	−0.04395	−0.0388	−0.07748	−0.09884	−0.123	−0.159	−0.184	−0.209	−0.227	−0.242	−0.256	−0.273	−0.285	−0.307	−0.315		
	0.167	0.027889	−0.327	−0.281	−0.16	−0.05473	−0.374	−0.319	−0.206	−0.138	−0.03637	−0.04403	−0.03894	−0.07786	−0.09946	−0.123	−0.16	−0.184	−0.209	−0.227	−0.243	−0.257	−0.273	−0.285	−0.308	−0.315		
	0.2	0.04	−0.327	−0.282	−0.16	−0.05486	−0.374	−0.32	−0.206	−0.139	−0.03724	−0.04412	−0.03916	−0.07829	−0.1	−0.124	−0.16	−0.185	−0.21	−0.228	−0.243	−0.257	−0.273	−0.285	−0.309	−0.315		
	0.234	0.054756	−0.328	−0.283	−0.16	−0.05504	−0.375	−0.32	−0.207	−0.14	−0.03849	−0.04428	−0.03981	−0.07905	−0.101	−0.125	−0.161	−0.185	−0.211	−0.23	−0.243	−0.259	−0.274	−0.285	−0.311	−0.316		
	0.267	0.071289	−0.329	−0.284	−0.161	−0.0553	−0.376	−0.321	−0.207	−0.141	−0.03979	−0.04437	−0.04083	−0.08039	−0.103	−0.126	−0.161	−0.186	−0.212	−0.231	−0.243	−0.26	−0.275	−0.286	−0.313	−0.316		
	0.3	0.09	−0.33	−0.285	−0.161	−0.05561	−0.376	−0.322	−0.208	−0.143	−0.04149	−0.04467	−0.042	−0.08163	−0.104	−0.127	−0.163	−0.188	−0.213	−0.233	−0.243	−0.262	−0.276	−0.286	−0.315	−0.317		
	0.334	0.111556	−0.331	−0.287	−0.162	−0.05618	−0.378	−0.323	−0.21	−0.145	−0.04329	−0.04497	−0.04363	−0.0831	−0.106	−0.129	−0.164	−0.189	−0.215	−0.235	−0.243	−0.264	−0.277	−0.287	−0.318	−0.317		
	0.367	0.134689	−0.333	−0.289	−0.162	−0.05666	−0.379	−0.324	−0.211	−0.147	−0.04564	−0.04537	−0.04531	−0.08502	−0.108	−0.13	−0.165	−0.191	−0.217	−0.237	−0.243	−0.266	−0.278	−0.288	−0.32	−0.318		
	0.4	0.16	−0.335	−0.291	−0.163	−0.0574	−0.381	−0.326	−0.213	−0.149	−0.04789	−0.04599	−0.04746	−0.08685	−0.11	−0.133	−0.167	−0.193	−0.219	−0.239	−0.244	−0.268	−0.28	−0.29	−0.323	−0.319		
	0.434	0.188356	−0.337	−0.293	−0.164	−0.05826	−0.383	−0.328	−0.215	−0.151	−0.05054	−0.04673	−0.04968	−0.08928	−0.113	−0.135	−0.17	−0.195	−0.221	−0.241	−0.245	−0.27	−0.282	−0.292	−0.325	−0.32		
	0.467	0.218089	−0.339	−0.296	−0.166	−0.05947	−0.385	−0.33	−0.217	−0.154	−0.0533	−0.04768	−0.05173	−0.09203	−0.116	−0.138	−0.172	−0.197	−0.224	−0.244	−0.246	−0.273	−0.284	−0.294	−0.329	−0.321		
	0.5	0.25	−0.342	−0.299	−0.167	−0.06072	−0.388	−0.332	−0.22	−0.157	−0.0564	−0.04935	−0.05457	−0.09528	−0.119	−0.14	−0.175	−0.2	−0.226	−0.247	−0.247	−0.275	−0.287	−0.296	−0.332	−0.322		
	0.534	0.285156	−0.345	−0.302	−0.169	−0.06203	−0.39	−0.335	−0.222	−0.16	−0.05985	−0.05066	−0.05751	−0.09857	−0.122	−0.144	−0.178	−0.203	−0.23	−0.251	−0.249	−0.278	−0.289	−0.299	−0.336	−0.324		
	0.567	0.321489	−0.348	−0.305	−0.171	−0.06342	−0.393	−0.338	−0.226	−0.164	−0.0634	−0.05253	−0.06046	−0.102	−0.126	−0.147	−0.181	−0.206	−0.233	−0.255	−0.251	−0.282	−0.292	−0.301	−0.34	−0.326		
	0.601	0.361201	−0.351	−0.309	−0.173	−0.06523	−0.396	−0.341	−0.229	−0.167	−0.06736	−0.05484	−0.06391	−0.106	−0.129	−0.151	−0.184	−0.209	−0.237	−0.258	−0.253	−0.285	−0.296	−0.305	−0.345	−0.328		
	0.634	0.401956	−0.355	−0.313	−0.176	−0.06714	−0.399	−0.344	−0.232	−0.171	−0.07153	−0.05725	−0.06758	−0.11	−0.134	−0.155	−0.188	−0.213	−0.24	−0.263	−0.255	−0.289	−0.299	−0.308	−0.349	−0.33		
	0.667	0.444889	−0.359	−0.317	−0.179	−0.06826	−0.403	−0.348	−0.236	−0.176	−0.07589	−0.06004	−0.07119	−0.114	−0.138	−0.159	−0.191	−0.217	−0.245	−0.267	−0.257	−0.293	−0.303	−0.312	−0.354	−0.333		
	0.701	0.491401	−0.363	−0.322	−0.182	−0.07082	−0.407	−0.352	−0.24	−0.18	−0.08049	−0.06281	−0.07558	−0.118	−0.142	−0.163	−0.196	−0.222	−0.249	−0.272	−0.26	−0.298	−0.306	−0.314	−0.359	−0.336		
	0.734	0.538756	−0.367	−0.326	−0.185	−0.07351	−0.411	−0.355	−0.244	−0.185	−0.08567	−0.0659	−0.08002	−0.123	−0.147	−0.167	−0.2	−0.227	−0.254	−0.277	−0.263	−0.302	−0.311	−0.319	−0.364	−0.339		
	0.767	0.588289	−0.372	−0.331	−0.188	−0.07663	−0.415	−0.36	−0.248	−0.19	−0.09098	−0.06927	−0.08478	−0.128	−0.152	−0.172	−0.205	−0.232	−0.259	−0.282	−0.267	−0.308	−0.315	−0.323	−0.369	−0.342		
	0.801	0.641601	−0.377	−0.336	−0.192	−0.07973	−0.42	−0.364	−0.253	−0.195	−0.09624	−0.07304	−0.08972	−0.133	−0.157	−0.177	−0.209	−0.237	−0.264	−0.288	−0.27	−0.313	−0.32	−0.327	−0.375	−0.345		
	0.834	0.695556	−0.382	−0.342	−0.196	−0.08327	−0.424	−0.369	−0.258	−0.2	−0.102	−0.07695	−0.09501	−0.139	−0.163	−0.182	−0.214	−0.242	−0.269	−0.293	−0.274	−0.318	−0.325	−0.332	−0.381	−0.349		
	0.868	0.753424	−0.387	−0.347	−0.201	−0.08676	−0.43	−0.374	−0.263	−0.206	−0.108	−0.08112	−0.101	−0.144	−0.169	−0.188	−0.22	−0.247	−0.274	−0.299	−0.278	−0.324	−0.33	−0.337	−0.387	−0.353		
	0.901	0.811801	−0.393	−0.354	−0.205	−0.09091	−0.435	−0.379	−0.269	−0.212	−0.114	−0.08542	−0.106	−0.15	−0.175	−0.194	−0.226	−0.253	−0.28	−0.305	−0.282	−0.33	−0.335	−0.342	−0.394	−0.357		
	0.934	0.872356	−0.399	−0.36	−0.21	−0.09495	−0.44	−0.385	−0.274	−0.218	−0.12	−0.09016	−0.112	−0.156	−0.181	−0.2	−0.232	−0.259	−0.287	−0.311	−0.287	−0.336	−0.341	−0.348	−0.4	−0.362		
	0.968	0.937024	−0.405	−0.366	−0.215	−0.09954		−0.39	−0.28	−0.225	−0.127	−0.09532	−0.118	−0.162	−0.188	−0.206	−0.237	−0.265	−0.293	−0.318	−0.291	−0.342	−0.347	−0.353	−0.407	−0.367		
	1.001	1.002001	−0.411	−0.372	−0.221	−0.104		−0.396	−0.286	−0.231	−0.134	−0.101	−0.125	−0.169	−0.194	−0.213	−0.244	−0.271	−0.299	−0.324	−0.296	−0.349	−0.353	−0.359	−0.414	−0.372		
	1.034	1.069156	−0.418	−0.379	−0.226	−0.109		−0.402	−0.293	−0.238	−0.141	−0.106	−0.132	−0.176	−0.201	−0.219	−0.25	−0.278	−0.306	−0.331	−0.302	−0.356	−0.359	−0.365	−0.421	−0.377		
	1.068	1.140624	−0.425	−0.387	−0.232	−0.114		−0.409	−0.299	−0.245	−0.149	−0.112	−0.138	−0.183	−0.208	−0.226	−0.257	−0.285	−0.313	−0.339	−0.307	−0.362	−0.365	−0.372	−0.429	−0.383		
	1.101	1.212201	−0.432	−0.395	−0.238	−0.12		−0.416	−0.306	−0.252	−0.157	−0.117	−0.145	−0.19	−0.216	−0.234	−0.264	−0.292	−0.32	−0.346	−0.313	−0.369	−0.372	−0.379	−0.436	−0.389		
	1.134	1.285956	−0.439	−0.402	−0.244	−0.125		−0.422	−0.314	−0.26	−0.165	−0.124	−0.153	−0.198	−0.224	−0.241	−0.271	−0.3	−0.328	−0.354	−0.319	−0.377	−0.379	−0.386	−0.43	−0.395		
	1.168	1.364224		−0.409	−0.25	−0.132		−0.43	−0.321	−0.268	−0.172	−0.13	−0.16	−0.205	−0.232	−0.248	−0.279	−0.307	−0.335	−0.362	−0.325	−0.385	−0.386	−0.393		−0.401		
	1.201	1.442401		−0.417	−0.257	−0.138		−0.437	−0.328	−0.276	−0.181	−0.137	−0.168	−0.213	−0.239	−0.256	−0.287	−0.315	−0.343	−0.37	−0.332	−0.393	−0.394	−0.4		−0.407		
	1.235	1.525225		−0.426	−0.265	−0.144			−0.336	−0.284	−0.189	−0.144	−0.176	−0.221	−0.248	−0.264	−0.295	−0.323	−0.351	−0.378	−0.339	−0.401	−0.402	−0.408		−0.413		
	1.268	1.607824		−0.434	−0.272	−0.15			−0.344	−0.292	−0.198	−0.151	−0.184	−0.23	−0.256	−0.272	−0.303	−0.331	−0.36	−0.387	−0.345	−0.409	−0.409	−0.415		−0.421		
	1.301	1.692601			−0.279	−0.157			−0.352	−0.301	−0.207	−0.159	−0.193	−0.239	−0.266	−0.28	−0.311	−0.341	−0.368	−0.396	−0.353	−0.418	−0.418	−0.423				

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